Tutorial – 07

Department of Physics **Shiv Nadar University**

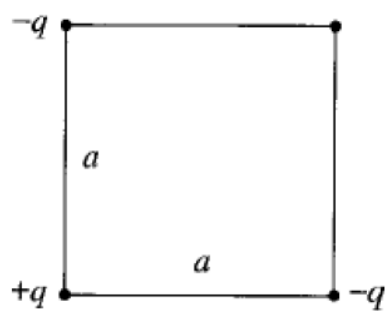
1.

Consider problem 1 of Tutorial 6: The volume charge density of a solid sphere of radius R varies as $\rho = \rho_0 \left(\frac{r}{R}\right)$, where ρ_0 is a constant (of appropriate unit) and r is the radial distance measured from the center of the sphere. Find out the electric field at a distance $'s'$ from the center of the sphere, this time ***with the help of Gauss's law***. Consider both $0 \leq s \leq R$ and $s \geq R$ cases.

[Compare the effort required to solve this problem by integration-approach (tutorial 6) and by Gauss's law. It should make you feel the immense power of Gauss's law when a good symmetry is present in the problem.]

2.

(a) Three charges are situated at the corners of a square (of side a) as shown. How much work does it take to bring in another charge, +q, from far away and place it in the fourth corner?



(b) How much work does it take to assemble the whole configuration of four charges?

3.

Find the energy stored in a uniformly charged solid sphere (volume charge density ρ) of radius R and total charge q by first calculating the potential (V) inside the sphere and then using the formula for the energy, $W = \frac{1}{2}\int_{\nu} \rho V d\tau$, where the region of integration ν is the volume occupied by the charges.

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4. Prove that the boundary conditions for electric field are satisfied for the case of

(i) an infinite plane sheet of charge with surface charge density σ, and (ii) a hollow cylindrical tube of radius R and uniform surface charge density σ.

[Hint: First find the electric field inside and outside of the two systems, then calculate the field at the surface and after, take the difference of the electric fields **just** inside and **just** outside the surface. The difference should be $\frac{\sigma}{\epsilon_0}$.]